Assignment No. 6

Name: Shubham Shivaji Phad
Class & Division: LY AIA 03 Batch B
Roll-No:48
Enrollment Number: ADT23SOCB1106

Title

Build a Named Entity Recognition (NER) application using SpaCy to automatically identify and classify entities such as persons, organizations, locations, dates, and monetary values from text.

Aim

To implement a Named Entity Recognition (NER) application using SpaCy to identify and classify entities such as persons, organizations, locations, dates, and monetary values from text.

Objectives

- Understand the concept of Named Entity Recognition.
- Use SpaCy's pretrained English model for NER.
- Identify and classify common named entities from text.
- Display extracted entities with their labels.

Theory

Named Entity Recognition is an NLP task that identifies important real-world entities in text and assigns them predefined labels. Common labels include **PERSON** for people, **ORG** for organizations, **GPE** for locations, **DATE** for dates, and **MONEY** for monetary values.

SpaCy is an industrial-strength NLP library that provides tokenization, part-of-speech tagging, dependency parsing, and named entity recognition. Its pretrained English model can recognize entities directly from natural language text.

Software/Libraries Required

- Python 3.x

- Google Colab / Jupyter Notebook
- SpaCy
- English model: `en_core_web_sm`

Installation Commands

Run the following commands in Google Colab:

```
!pip install spacy -q
!python -m spacy download en_core_web_sm -q
```

```
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✔ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting
the
'Restart kernel' or 'Restart runtime' option.
```

Algorithm/Steps

1. Install SpaCy and download the English language model.
2. Import SpaCy and load `en_core_web_sm`.
3. Define the input text containing different types of entities.
4. Process the text using the SpaCy NLP pipeline.
5. Access the detected entities through `doc.ents`.
6. Display each entity and its corresponding label.

Program with Output

```
import spacy

nlp = spacy.load("en_core_web_sm")

text = (
    "John Smith joined Microsoft in New York on 15 March 2026. "
    "The company announced an investment of $25 million."
)

doc = nlp(text)

print("Named Entities:")
for ent in doc.ents:
    print(f"{ent.text} -> {ent.label_}")
```

```
Named Entities:
John Smith -> PERSON
Microsoft -> ORG
New York -> GPE
15 March 2026 -> DATE
$25 million -> MONEY
```

Output:

```
Named Entities:
John Smith -> PERSON
Microsoft -> ORG
New York -> GPE
15 March 2026 -> DATE
$25 million -> MONEY
```

The exact entity detection may vary slightly with the SpaCy model version.

Conclusion

The Named Entity Recognition application was successfully implemented using SpaCy. The system identified and classified entities such as persons, organizations, locations, dates, and monetary values from the given text. NER is useful for information extraction and other NLP applications.